QPCODE

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) CHEMISTRY

THIRD SEMESTER EXAMINATION- MODEL QUESTION PAPER

DSC A - MG3DSCCHE201- ORGANIC CHEMISTRY-1

(2024 ADMISION ONWARDSADMISSION ONWARDS)

Duration: 1.5 hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

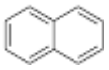
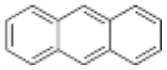
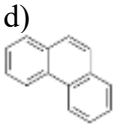
Answer all questions

Each question carries **1** marks

1. What is the main purpose of cracking in the petroleum industry? [K] / [1]
   A) To remove sulphur from crude oil
   B) To convert solid fuel into liquid fuel
   C) To increase the yield of gasoline and other useful products
   D) To extract pure methane from natural gas

2. Which of the following reagents is essential for the Wurtz reaction? [U] / [1]
   A) Zinc and hydrochloric acid
   B) Sodium metal in dry ether
   C) Potassium hydroxide in ethanol
   D) Concentrated sulfuric acid

3. Which among the following compounds is anthracene? [U] / [2]

a)  b)  c)  d) 

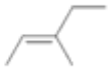
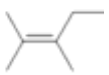
4. What is the resonance energy of benzene [K] / [2]

   a) 150.6 kcal/mol b) 36 kcal/mol c) 61 kcal/mol d) 84 kcal/mol

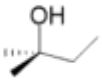
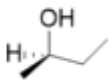
5. What is the monomer of polythene? [K] / [1]

   a) vinyl chloride b) ethylene c) propylene d) LDL

6. Which of the following compounds can exhibit geometrical isomerism. [An] / [3]

a)  b)  c)  d) 

7. Analyse the structures of the following compounds and select the one that is chiral. [An] / [3]

a)  b)  c)  d) 

8. Which among the following exist as gas at normal temperature and pressure? [K] / [1]

   a) acetylene b) octane c) tartaric acid d) polythene

9. Which of the following compounds is the most acidic? [U] / [1]

   a) ethane b) ethene c) ethyne d) methane

10. Which among the following is an anti-aromatic compound? [A] / [2]

    a) cyclopentene b) cyclopropene c) cyclobutadiene d) cyclopentadiene

[1x10 = 10]

Part B

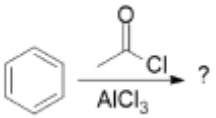
Short Answer Questions
Answer any **4** questions
Each question carries **3** marks

11. Draw the mechanism of bromination of ethylene. [U] / [1]

12. Explain Huckel rule. [U] / [2]

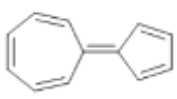
13. What is Lindlar's catalyst? Explain its composition and use in organic synthesis. Give one example of a reaction where it is used. [U] / [1]

14. Illustrate the mechanism of the following reaction and predict the product. [A] / [2]



15. Differentiate between D/L notation and d/l notation [U] / [3]

16. Explain why the given hydrocarbon exhibits polarity, using the concept of aromaticity to support your reasoning. [An / [2]



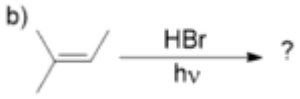
[3x4 = 12]

Part C

Short Essay Questions
Answer any 2 questions
Each question carries **7** marks

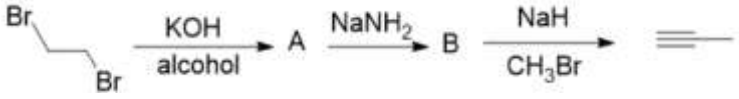
17. Predict the possible products of the following reaction and justify which would be the major and minor products, supporting your answer with the appropriate reaction mechanisms. [E] / [1]

a) 

b) 

18 Draw the Newman projections for major conformations of n-butane about the C2–C3 bond. Explain their relative stabilities and sketch the energy profile for one full rotation. [U] / [3]

19. Predict the structures of products A and B in the given reaction scheme, and explain in detail the reasoning and steps you used to arrive at your answer. [E] / [1]

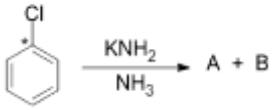


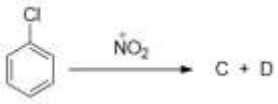
[2x7 = 14]

Part D

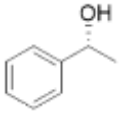
Essay Question
Answer any 1 question
Each question carries 14 marks

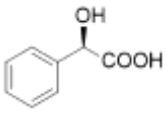
20 Predict the products A, B, C and D. Reason your answer. [An] / [2]

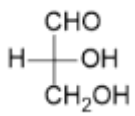
a) 

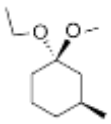
b) 

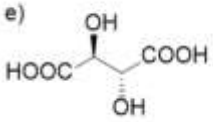
21 For each of the following molecules, predict whether it is chiral or achiral. For molecules containing chiral centres, assign the R and S configuration(s) to each chiral centre. Provide a detailed, step-by-step explanation of how you determined chirality and stereochemical configuration. [An] / [3]

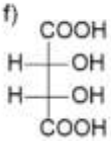
a) 

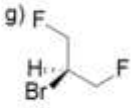
b) 

c) 

d) 

e) 

f) 

g) 

[1x14 = 14]